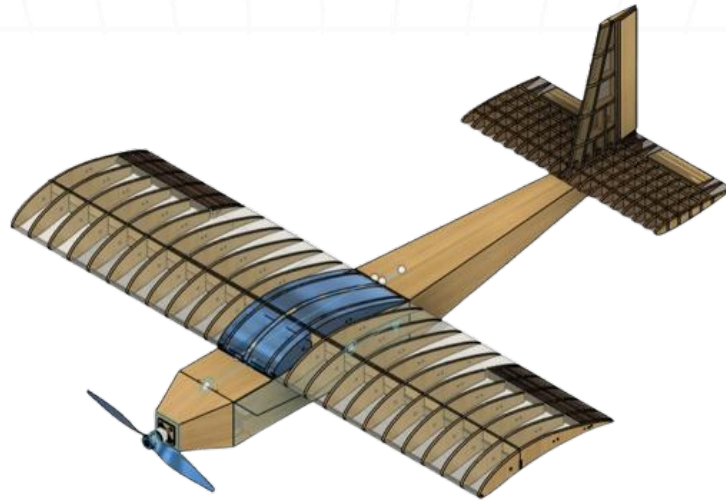




Boeing National Aeromodelling Competition 2024-25



Team After Burners

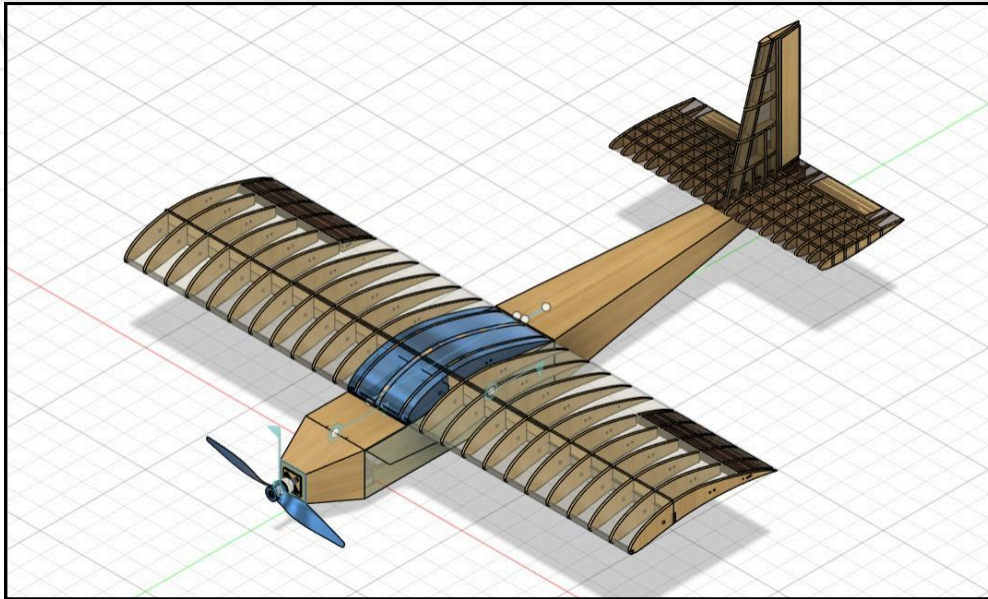
Aeromodelling Club, IIT Guwahati

IIT Kharagpur Zone

Team Details

- Pranav Shalya
- Harshit Goel
- Sumit Singhatia
- Anuraj Gogoi

1.1 Design



1.2 Dimensions and Performance Parameters

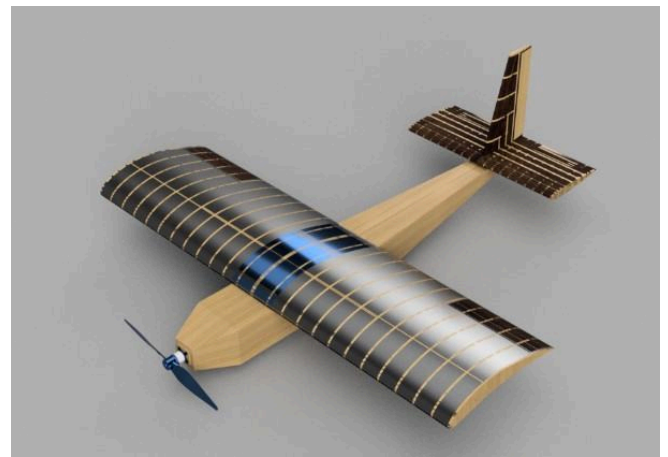
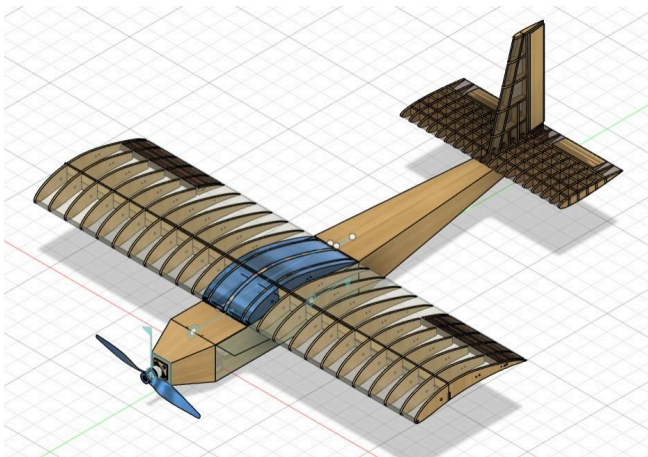
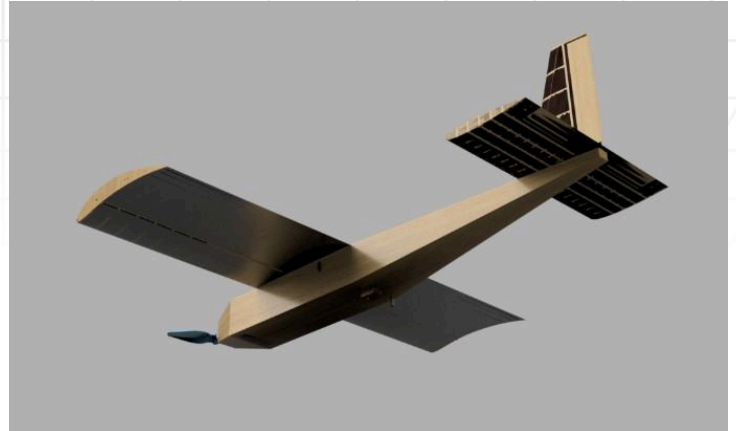
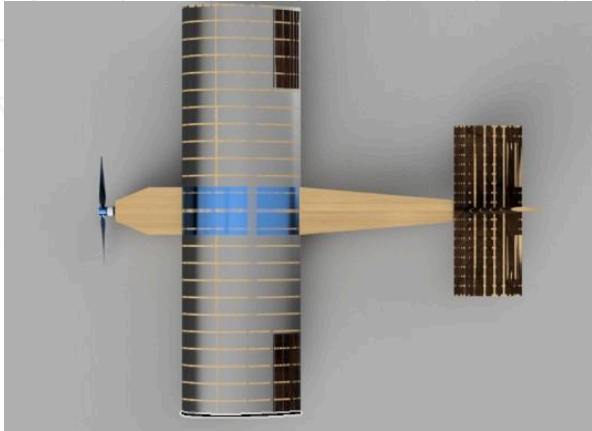
Conditions	
Air Speed	13 m/s
Air Density	1.225 kg/m ³

Weight	
Airplane (without payload)	1.1 kg
Payload	1 kg
Total	2.1 kg

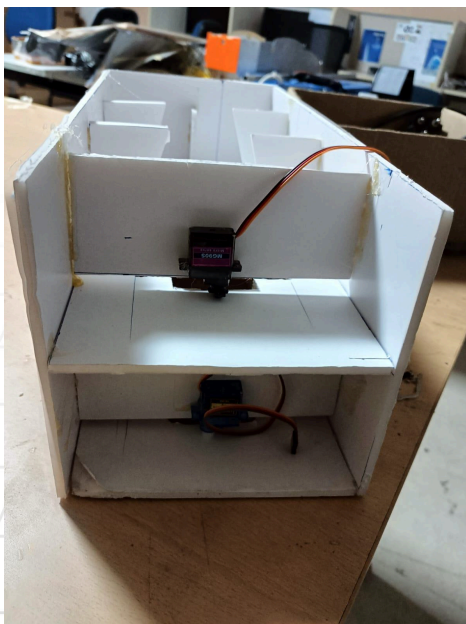
Main Wing	
Wing Loading	4.38 kg/m ²
Aspect Ratio	3
Wing Area	0.48 m ²
Root Chord	0.4 m
Tip Chord	0.4 m
Taper Ratio	1

1.3 Machine Drawing Of The Aircraft

Aircraft Design

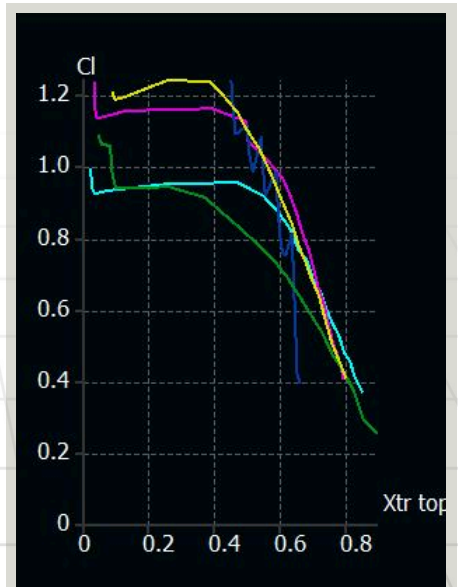
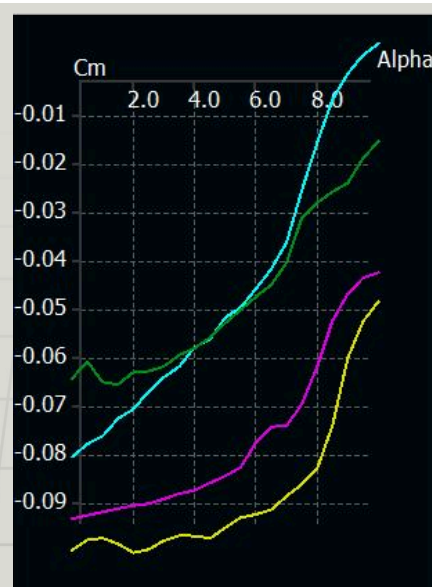
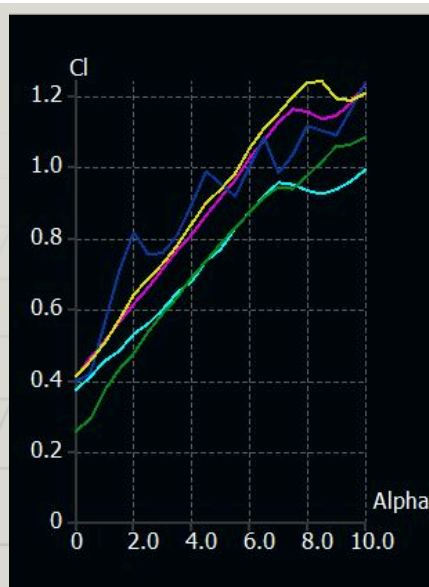
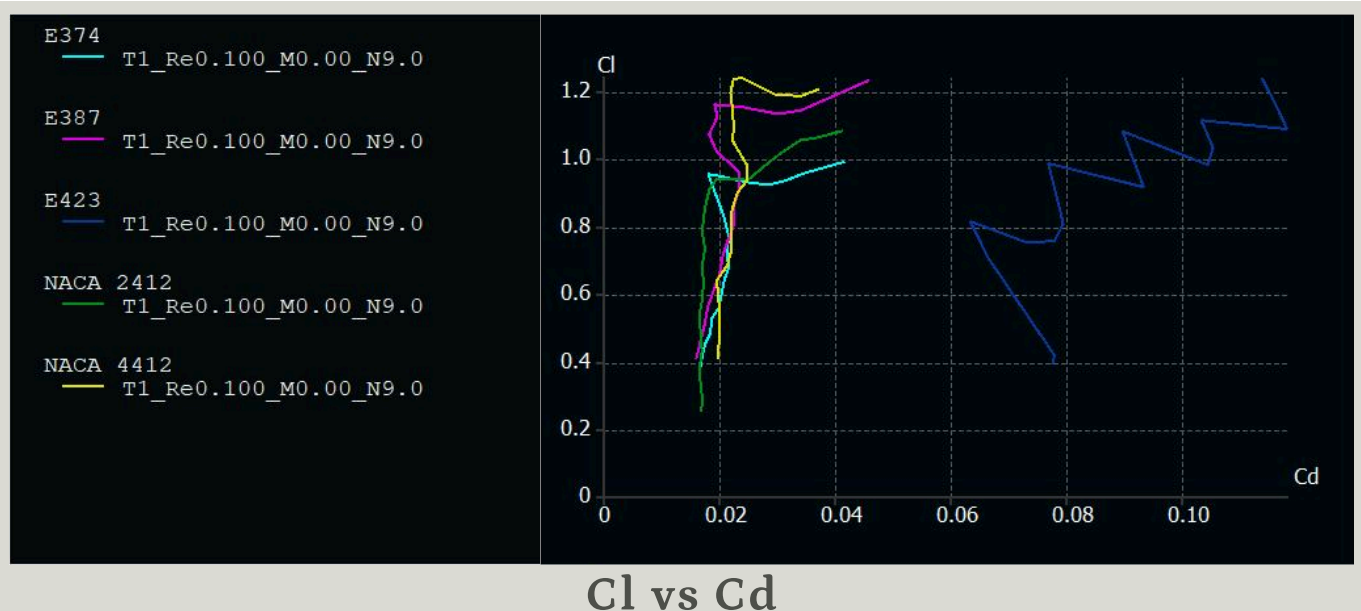


Dropping Mechanism



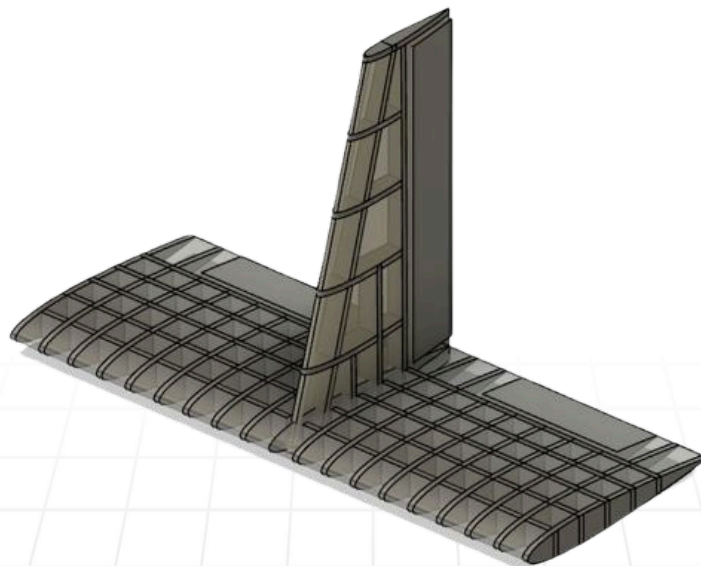
1.4 Wing Design And Specifications

The wing design prioritizes maximizing payload, necessitating a focus on lift generation. A high-wing configuration with a rectangular cross-section was chosen, featuring a 1.2-meter wingspan and a root chord of 0.40m and a tip chord of 0.40. To achieve this, a comparative analysis of airfoil profiles was conducted. While the e423 airfoil exhibited slightly lower lift-to-drag ratios compared to some alternatives, it demonstrated superior lift generation capabilities. Recognizing that a slight compromise in efficiency was acceptable to achieve the desired payload capacity, the e423 airfoil was selected for its ability to provide the highest lift.

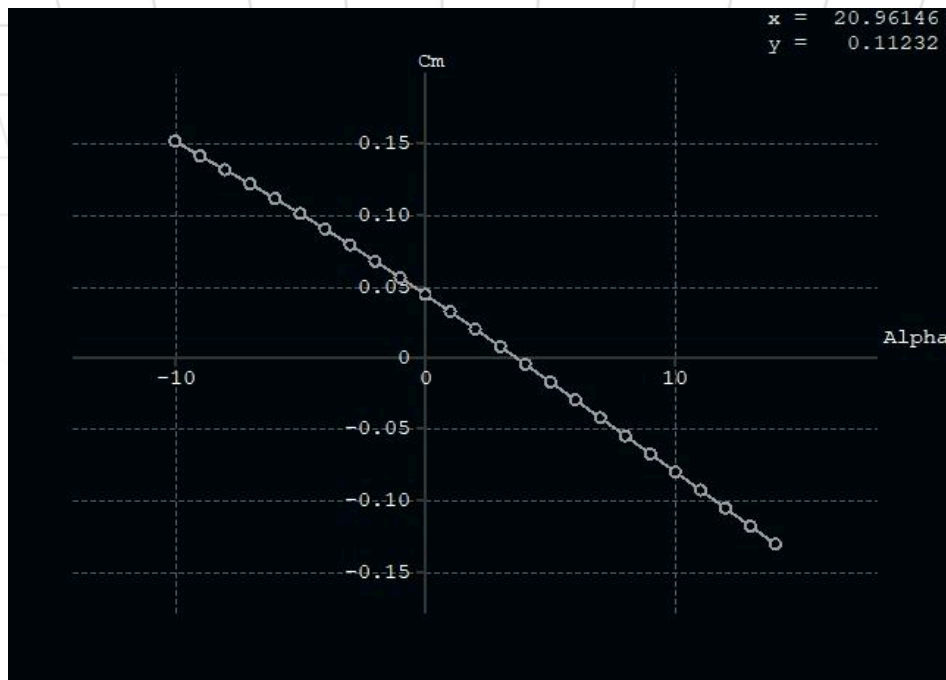


1.5 TAIL OF THE WING

- We evaluated different tail configurations including symmetrical airfoils, cambered airfoils, and flat plates. Each configuration has its own set of advantages and disadvantages. While more complex shapes might offer slight aerodynamic advantages in specific situations, the benefits of simplicity and ease of manufacturing often make the rectangular configuration a compelling choice for many aircraft designs.
- The horizontal tail design targeted an ideal aspect ratio between 3 and 5. To optimize lift-to-drag ratio (CL/CD) and angle of attack (α) for zero pitching moment (0 CM), the team selected a span length of 50 cm and a tail area of 1100 cm².
- The vertical tail design prioritized stability by targeting a smaller aspect ratio between 1 and 2. The team optimized the span area and chord length, resulting in a final aspect ratio of 1.86. capacity, the e423 airfoil was selected for its ability to provide the highest lift.



1.6 STABILITY ANALYSIS



Zero lift pitching moment = 0.14
static margin = 15%

Stability analysis was conducted using XFLR5 on the full aircraft model to assess longitudinal static stability. The following results were obtained.

- The plot of Moment Coefficient (C_m) versus Angle of Attack (α) exhibits a negative linear slope, indicating a longitudinally statically stable aircraft.
- The slope of the C_m - α curve confirms that an increase in angle of attack produces a restoring (nose-down) moment, promoting self-correction in pitch.
- The moment coefficient at zero angle of attack (C_{m_0}) is slightly negative, which is consistent with the presence of a stabilizing horizontal tail generating a downward force.

1.7 PROPULSION SYSTEM

Components	
Battery	4s 1800 mAh (30C)
ESC	40A
Motor	830 kv
Propeller	12x6
Power	~470-500W
Current Draw	~2.2-2.5 kg

1.7.1 Wattage

Since our RC Plane is to be designed with a decent payload carrying capacity but we do not need a very high flying speed, we can take the power performance level to be somewhere between basic trainer and sport flying rc plane.

Hence we take the power performance level of our plane to be 100 watts/pound or **222.2 watts/kg**.

$$\begin{aligned}\text{Wattage} &= 2.1\text{kg} \times 222.2 \text{ W/kg} \\ \text{Wattage} &\approx 466.6 \text{ W} \\ \text{Wattage Required:} &\approx 467 \text{ W}\end{aligned}$$

1.7.2 Current

- 1 cell nominal voltage: 3.7V
- 4 cells → 14.8V nominal

$$\text{Current} = \text{Voltage} / \text{Wattage} = 14.8 / 466.6 \approx 31.5\text{A}$$

Add 20% headroom for ESC:

$$\text{ESC Current} \approx 31.5 \times 1.2 \approx 38\text{A}$$

ESC: 40A (4S compatible)

1.7.3 Battery C Rating

Battery capacity = 1800 mAh = 1.8 Ah

C rating required:

$$C = \text{Max Current} \times \text{Capacity} \times 2 = 35 / 1.8 \times 2 \approx 38.9\text{C}$$

4S 1800 mAh 40C battery minimum.

1.7.4 Motor-Propeller-Battery Configuration Analysis

General	Model Weight: 2100 g incl. Drive 74.1 oz	# of Motors: 1 (on same Battery)	Wingspan: 1200 mm 47.24 inch	Wing Area: 48 dm² 744 in²	Drag: simplified 0.03 Cd	Field Elevation: 500 m ASL 1640 ft ASL	Air Temperature: 25 °C 77 °F	Pressure (QNH): 1013 hPa 29.91 inHg
Battery Cell	Type (Cont. / max. C) - charge state: LiPo 1800mAh - 30/45C - normal	Configuration: 4 S 1 P	Cell Capacity: 1800 mAh 1800 mAh total	max. discharge: 85%	Resistance: 0.0094 Ohm	Voltage: 3.7 V	C-Rate: 30 C cont. 45 C max	Weight: 46 g 1.6 oz
Controller	Type - Timing: max 50A - normal	Current: 50 A cont. 50 A max	Resistance: 0.005 Ohm	Weight: 65 g 2.3 oz	Battery extension Wire: AWG10=5.27mm²	Length: 0 mm 0 inch	Motor extension Wire: AWG10=5.27mm²	Length: 0 mm 0 inch
Motor	Manufacturer - Type (Kv) - Cooling: (* = discontinued) EMAX - EcoII 2814-830 (830) - medium	KV (w/o torque): 830 rpm/V Prop-Kv-Wizard	no-load Current: 1.3 A @ 10 V	Limit (up to 15s): 1580 W	Resistance: 0.064 Ohm	Case Length: 29 mm 1.14 inch	# mag. Poles: 14	Weight: 89 g 3.1 oz
Propeller	Type - yoke twist: Generic - normal - 0°	Diameter: 12 inch 304.8 mm	Pitch: 6 inch 152.4 mm	# Blades: 2	PConst / TConst: 1.07 / 1.0	Gear Ratio: 1 : 1	Flight Speed: 0 km/h 0 mph	calculate

Load: 19.0

Mixed Flight Time: 4.3

electric Power: 456

est. Temperature: 89

Thrust-Weight: 1.01

Pitch Speed: 80

Conclusion

The selected setup (4S 1800 mAh, 800 KV motor, 12×6 prop, 40A ESC) delivers 467 W power, a 1.01 thrust-to-weight ratio, and ensures efficient operation with only 18% motor load.

This configuration is ideal for stable short-duration cargo missions, providing reliable performance and safe operation.

1.8 Results

Our heavy lifter design incorporates the e423 airfoil for optimal lift generation and features lightweight stabilizers for stability and ease of construction. Calculations, including adherence to relevant regulations, have been conducted to ensure the aircraft's safe and efficient operation while meeting the competition's specific objectives.

- We picked the e423 high-lift airfoil because, after comparing options, it seemed like the best choice for maximizing lift. For the tail configurations, we went for lightweight stabilizers to make things stable and easy to build. We backed this decision with stability tests and graphs, taking a scientific route to ensure the plane stays steady in the air.
- Through calculations and considering various constraints and parameters we analyze the aircraft and verified that it does not violates any law on paper
- To sum it up, our aircraft design perfectly fits the goals and objectives of the competition.

1.9 References

- **TAIL CALCULATION-** (References: aircraft_design-libre.pdf (d1wqtxts1xzle7.cloudfront.net))
(Department of Aerospace Engineering, IIT MADRAS)
- **PROPULSION SYSTEM–** Selecting a Brushless Motor, ESC, LiPo Battery and Prop for an RC Airplane
- **THRUST-** (Reference : J.D Anderson “Introduction to Flight “ 8th edition)
- **DRAG-** Reference : Lee_Nicholai's_White_Paper
: https://youtu.be/QLGW3a_U6hY
: Nicolai, Leland M., “Fundamentals of Aircraft Design”, METS Inc, 6520 Kingsland Ct, San Jose, Ca
- **PERFORMANCE-** (Reference: Nicolai, Leland M., “Fundamentals of Aircraft Design”, METS Inc, 6520 Kingsland Ct, San Jose, Ca 95120, 1975)